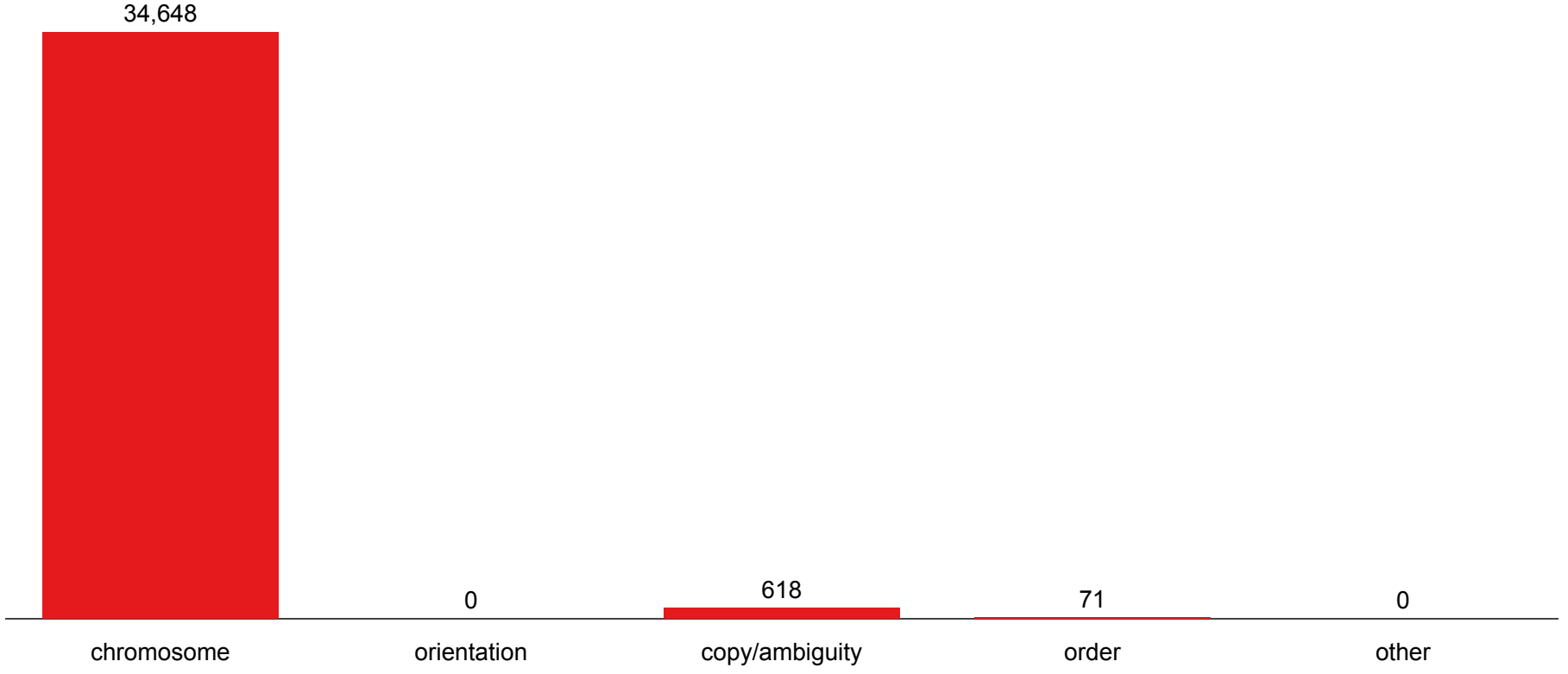Primary remaining boundary causes